

Chapter 2—Mendel’s Breakthrough: Patterns, Particles, and Principles of Heredity

Fill in the Blank

1. A _____ is a sequence of nucleotides that are basic units of biological information.
Ans: gene
Difficulty: 1
2. _____ is the science of heredity and it seeks a precise explanation of the biological structures and mechanisms that determine what is inherited and how it is inherited.
Ans: Genetics
Difficulty: 1
3. _____ is the purposeful control over mating by choice of parents for the next generation.
Ans: Artificial selection
Difficulty: 1
4. _____ is the process whereby both egg and pollen come from the same plant.
Ans: Self fertilization
Difficulty: 1
5. _____ is a Mendelian law that states both alleles must separate during gamete formation.
Ans: Law of segregation
Difficulty: 1
6. _____, _____ and _____ all were involved in the rediscovery of Mendel's research.
Ans: Corens, deVaries, Tschermak
Difficulty: 1

Describe below the meaning of each symbol.

7.



Ans: Normal male

Difficulty: 1

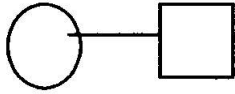
8.



Ans: Normal female

Difficulty: 1

9.



Ans: Marriage

Difficulty: 1

10.



Ans: Affected male

Difficulty: 2

11. _____ Inherited trait expressed only when the controlling gene is homozygous.

Ans: Recessive

Difficulty: 2

12. _____ One contrasting form of a gene.

Ans: Allele

Difficulty: 2

13. _____ Symbol for the grandparents of a given individual.

Ans: P2

Difficulty: 2

14. _____ The observable properties of an organism that are under genetic control.

Ans: Phenotype

Difficulty: 2

Multiple Choice

15. The first offspring from the parents are called:

- A) P.
- B) F₁.
- C) F₂.
- D) testcross.
- E) backcross.

Ans: B

Difficulty: 1

16. A _____ is a cross between an unknown and a homozygous recessive.

- A) testcross
- B) dihybrid
- C) monohybrid
- D) backcross
- E) controlled

Ans: A

Difficulty: 1

17. If an individual has 10 gene pairs, how many different gametes can be formed if three of the gene pairs are homozygous and the remaining 7 gene pairs are heterozygous?

- A) 49
- B) 100
- C) 128
- D) 1024
- E) 131,072

Ans: C

Difficulty: 4

18. If the parents of a family already have two boys, what is the probability that the next two offspring will be girls?

- A) 1
- B) 1/2
- C) 1/3
- D) 1/4
- E) 1/8

Ans: D

Difficulty: 2

19. In some genetically engineered corn plants the dominant gene (BT) produces a protein that is lethal to certain flying insect pests that eat the corn plants. It was also found that the pollen could also cause death in some flying insects. If the corn plant is heterozygous for BT, what proportion of the pollen would carry the dominant gene?

- A) All pollen
- B) $\frac{1}{2}$
- C) $\frac{1}{3}$
- D) $\frac{1}{4}$
- E) $\frac{1}{8}$

Ans: B

Difficulty: 1

20. Suppose that in plants, smooth seeds (S) are dominant to wrinkle seeds (s) and tall plants (T) are dominant to short plants (t). A tall plant with smooth seeds was backcrossed to a parent that was short and wrinkle. What proportion of the progeny is expected to be heterozygous for tall and smooth?

- A) $\frac{1}{2}$
- B) $\frac{1}{4}$
- C) $\frac{1}{8}$
- D) $\frac{1}{16}$
- E) 0

Ans: B

Difficulty: 2

21. Suppose that in plants, smooth seeds (S) are dominant to wrinkle seeds (s) and tall plants (T) are dominant to short plants (t). A tall plant with smooth seeds was backcrossed to a parent that was short and wrinkle. What proportion of the progeny is expected to be homozygous for short and wrinkle?

- A) $\frac{1}{2}$
- B) $\frac{1}{4}$
- C) $\frac{1}{8}$
- D) $\frac{1}{16}$
- E) 0

Ans: B

Difficulty: 2

22. Sickle cell anemia is a recessive trait in humans. In a cross between a father who has sickle cell anemia and a mother is heterozygous for the gene, what is the probability that their first three children will be normal?

- A) 1/4
- B) 1/2
- C) none
- D) 1/8
- E) 1/16 will be albino

Ans: D

Difficulty: 2

23. The dominant trait, Huntington disease causes severe neural/brain damage at approximately age 40. A female whose mother has Huntington disease, marries a male whose parents are normal. It is not known if the female has the disease. What is the probability that their first-born will inherit the gene that causes Huntington disease?

- A) 25%
- B) 50%
- C) 75%
- D) 100%
- E) 0%

Ans: A

Difficulty: 1

24. In a monohybrid cross $AA \times aa$, how many homozygotes are found among the F_2 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: B

Difficulty: 1

25. In a monohybrid cross $AA \times aa$, how many heterozygotes are found among the F_2 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: B

Difficulty: 1

26. In a dihybrid cross $AAbb \times aaBB$, how many homozygotes are found among the F_2 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: D

Difficulty: 1

27. In a dihybrid cross $AABB \times aabb$, how many heterozygotes at both gene pairs are found among the F_2 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: D

Difficulty: 1

28. In the dihybrid cross $AaBb \times aabb$, how many homozygotes are found among the F_1 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: A

Difficulty: 1

29. In the dihybrid cross $AaBb \times aabb$, how many heterozygotes for both gene pairs are found among the F_1 offspring?

- A) one
- B) two
- C) three
- D) four
- E) five

Ans: A

Difficulty: 1

30. Among the dihybrid crosses below, which will give a 1:1 ratio?

- A) $AABB \times aabb$
- B) $AaBb \times AaBb$
- C) $AaBb \times aabb$
- D) $AaBB \times aaBB$
- E) $AAbb \times aaBB$

Ans: D

Difficulty: 1

31. Among the dihybrid crosses below, which will give a 1:1:1:1 ratio?

- A) $AABB \times aabb$
- B) $AaBb \times AaBb$
- C) $AaBb \times aabb$
- D) $AaBB \times aaBB$
- E) $AAbb \times aaBB$

Ans: B

Difficulty: 1

32. Assume that in guinea pigs, dark brown fur (B) is dominant to black fur (b). If you mate a heterozygous black guinea pig with a homozygous brown guinea pig, what proportion of the progeny will be black?

- A) none
- B) $1/4$
- C) $1/2$
- D) $3/4$
- E) all

Ans: D

Difficulty: 1

33. Assume that in guinea pigs, dark brown fur (B) is dominant to black fur (b). If you mate a homozygous black guinea pig with a homozygous brown guinea pig, what proportion of the progeny will be heterozygous?

- A) none
- B) $1/4$
- C) $1/2$
- D) $3/4$
- E) all

Ans: C

Difficulty: 1

34. Assume that in guinea pigs, dark brown fur (B) is dominant to black fur (b). If you mate a black guinea pig with a homozygous brown guinea pig, what proportion of the progeny will be homozygous?

- A) none
- B) 1/4
- C) 1/2
- D) 3/4
- E) all

Ans: C

Difficulty: 1

Matching

Match the following terms with the best definition

- a. self-fertilization
- b. cross fertilize
- c. monohybrid crosses
- d. artificial selection
- e. reciprocal crosses

35. _____ the purposeful control of mating by choice of parents for the next generation.

Ans: d

Difficulty: 1

36. _____ fertilization in which both egg and pollen come from the same plant, resulting in offspring with the same genetic traits as the single parent.

Ans: a

Difficulty: 1

37. _____ to brush the pollen from one plant onto the female organ of another plant, thereby creating offspring with the particular traits of the selected parent plants.

Ans: b

Difficulty: 1

38. _____ crosses in which the male and female traits are reversed, thereby controlling whether a particular trait is transmitted by the egg or the pollen.

Ans: e

Difficulty: 1

39. _____ crosses between parents that differ in only one trait.

- A) self-fertilization
- B) cross fertilize
- C) monohybrid crosses
- D) artificial selection
- E) reciprocal crosses

Ans: C

Difficulty: 1

40. An allele that expresses its phenotype even when heterozygous with a recessive allele is termed:

- A) recessive
- B) recombinant
- C) dominant
- D) parental
- E) independant

Ans: C

Difficulty: 1

41. An alternative form of a single gene is known as a(n):

- A) parental
- B) dihybrid
- C) reciprocal
- D) allele
- E) recessive

Ans: D

Difficulty: 1

42. The diploid cell formed by the fertilization of the egg by the sperm during sexual reproduction is a(n):

- A) reciprocal
- B) zygote
- C) dihybrid
- D) gamete
- E) monohybrids

Ans: B

Difficulty: 1

43. A phenotype reflecting a new combination of genes occurring during gamete formation is called:

- A) a recombinant type
- B) an independent assortment
- C) heterozygous
- D) homozygous
- E) a mutlihybrid cross

Ans: A

Difficulty: 1

44. The actual alleles present in an individual are called their:

- A) recombinant types
- B) zygote
- C) dominant allele
- D) allele
- E) genotype

Ans: E

Difficulty: 1

True or False

45. Mendel was the only botanist to work with large numbers of offspring, to count all offspring, subject his results to statistical analysis and then compare his results with predictions based on his models.

Ans: False

Difficulty: 2

46. The mating of parents with antagonistic traits produces hybrids.

Ans: True

Difficulty: 1

47. Mendel's law of segregation states that two alleles for each trait unite in a specific manner during gamete formation and therefore give rise to predictable observable traits.

Ans: False

Difficulty: 2

48. The Punnett square was introduced in 1906 by Reginald Punnett and provides a simple and convenient method of tracking possible combinations of gametes that might be produced in a given cross.

Ans: True

Difficulty: 1

49. Using the product rule, one would calculate the probability of parents having six children who are all boys as $(\frac{1}{2})^6$.
Ans: True
Difficulty: 1
50. The sum rule states that the probability of both of two mutually exclusive events occurring is the sum of their individual probabilities.
Ans: False
Difficulty: 3 (either, not both)
51. If you know the phenotype and the dominance relation of the alleles you can predict the genotype.
Ans: True
Difficulty: 1
52. An individual can be a heterozygote for one trait and a homozygote for another.
Ans: True
Difficulty: 1
53. A testcross is a cross between two heterozygotes.
Ans: False
Difficulty: 2
54. At fertilization, in the mating of dihybrids, four different kinds of eggs can combine with four different kinds of pollen, producing a total of 16 different genotypes.
Ans: False
Difficulty: 2
55. During gamete formation, different pairs of alleles segregate independently of each other.
Ans: False
Difficulty: 3 (not if linked)
56. If yellow and round phenotypes in peas are dominant, you know the genotype of all peas that are green and wrinkled.
Ans: False
Difficulty: 2
57. A pedigree is a family history of a specific trait shown for a minimum of three generations.
Ans: True
Difficulty: 1

58. Several single-gene disorders are more common in some populations of people than others.
Ans: True
Difficulty: 1
59. A lethal disorder does not include the inheritance of traits that cause death in adulthood.
Ans: False
Difficulty: 2
60. The following symbols $\sim = \pm$ indicate a consanguineous mating.
Ans: True
Difficulty: 1
61. Cross-fertilization is the same thing as reciprocal cross.
Ans: False
Difficulty: 2
62. The first filial generation is the offspring of parents.
Ans: True
Difficulty: 1
63. A zygote is a fertilized egg.
Ans: False
Difficulty: 1
64. A YY or yy genotype is called heterozygous.
Ans: False
Difficulty: 2
65. When Mendel repeated his pea experiments in bean, he found flowers that ranged from white to pale violet to purple. This is due to bean flower color being determined by more than one gene.
Ans: True
Difficulty: 1

Short Answer

66. In corn liguleless, (l^1) is recessive to ligules (L^1) and a green leaf (G) is dominant to the normal non-green (g). If a plant homozygous for liguleless and green leaves is crossed to one homozygous for non-green with ligules, predict the phenotypes and genotypes of the F1.
Ans:

<u>Genotype</u>	<u>Phenotype</u>
LLGg	Ligules/Green

Difficulty: 1

67. If an F1 plant from the question above were test crossed, what would be the phenotypes and genotypes of the progeny?

Ans:

<u>Genotype</u>	<u>Phenotype</u>
LlGg/Ligules	Green
Llgg/Ligules	non-green
llGg/Liguleless	Green
llgg/Liguleless	non-green

Difficulty: 3

68. In corn liguleless, (l^1) is recessive to ligules (L^1) and a green leaf (G) is dominant to the normal non-green (g). If a plant homozygous for liguleless and green leaves is crossed to one homozygous for non-green with ligules predict the phenotypes and genotypes of the F2.

Ans:

<u>Genotype</u>	<u>Phenotype</u>
1:LLGG/Ligules	Green
2:LLGg/Ligules	Green
2:LlGG/Ligules	Green
4:LlGg/Ligules	Green
1:LLgg/Ligules	non-green
2:Llgg/Ligules	non-green
1:llGG/Liguleless	Green
2:llGg/Liguleless	Green
1:llgg/Liguleless	non-green

Difficulty: 3

69. In *Drosophila*, forked (fk) bristles are recessive to normal (fk+) and glassy eyes (gls) are recessive to wild (gls+). If a homozygous wild-type male is mated to a forked bristle, glassy eye female, predict the genotypes and phenotypes of the F1.

Ans:

<u>Genotype</u>	<u>Phenotype</u>
fk+fk gls+gls	wildtype

Difficulty: 1

70. If an F1 female from the cross in question 69. was backcrossed to its male parent, predict the genotypes and phenotypes of the offspring.

Ans:

<u>Genotype</u>	<u>Phenotype</u>
fk ⁺ fk ⁺ gls ⁺ gls ⁺	Wild type
fk ⁺ fk ⁺ gls ⁺ gls	Wild type
fk ⁺ fk gls ⁺ gls ⁺	Wild type
fk ⁺ fk gls ⁺ gls	Wild type

Difficulty: 2

71. In *Drosophila*, forked (fk) bristles are recessive to normal (fk+) and glassy eyes (gls) are recessive to wild (gls+). If a homozygous wild-type male is mated to a forked bristle, glassy eye female, predict the genotypes and phenotypes of the F2.

Ans:

<u>Genotype</u>	<u>Phenotype</u>
1:fk+fk+ gls+gls+	Wild
2:fk+fk+ gls+gls	Wild
2:fk+fk gls+gls+	Wild
4:fk+fk gls+gls	Wild
1:fk+fk+ glsgls	Glassy
2:fk+fk glsgls	Glassy
1:fkfk gls+gls+	Forked
2:fkfk gls+gls	Forked
1:fkfkglsgls	Forked and Glassy

Difficulty: 3

72. A rosy-eyed *Drosophila* was crossed with a forked *Drosophila*. The F1 were all wild-type while the F2 consisted of 306 wild-type, 94 rosy-eyed, 102 fork-bristled and 33 forked-bristled and rosy-eyed flies. Infer the genotypes of the parents.

Ans: Both parents are homozygotes; AA^{bb} × aaBB.

Difficulty: 2

73. In pecans, the outer shell may be thick (T) or thin (t). The shell of pecans is the pericarp. If you use the pollen from a homozygous thick shell to pollinate a thin shell tree, what shell type would form on the tree following the cross?

Phenotype	Genotype

Ans: Genotype Phenotype
 pp thin

Difficulty: 3

74. If you use the seed from the pecans of the above cross to produce an F₂, what type of shells will form on the F₁ plant?

Phenotype	Genotype

Ans: Genotype Phenotype
 Pp thick

Difficulty: 3

75. After a cross between two corn plants, the F₁ plants all had a dwarfed phenotype. The F₂ consisted of 1,207 dwarf plants and 401 tall plants. Identify the phenotypes and genotypes of the two parents.

Parent A	Parent B
Genotype _____	Genotype _____
Phenotype _____	Phenotype _____

Ans: Parent A Parent B
 DD/dwarf dd/tall

Difficulty: 2

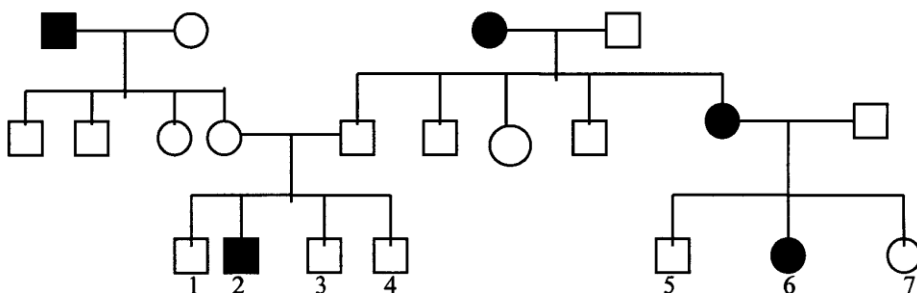
76. After a cross between two mice, the F₁ offspring all had the same phenotype. The F₂ consisted of 91 short tails and 29 normal tails. Identify the phenotypes and genotypes of the two parents plants.

Parent A	Parent B
Genotype _____	Genotype _____
Phenotype _____	Phenotype _____

Ans: Parent A Parent B
 SS/Short tails ss/Long tails

Difficulty: 2

Below is a pedigree for a human trait. Shaded symbols are for individuals showing the trait. (A) Identify the mode of inheritance of the trait. (B) Apply the laws of probability and calculate the probability the offspring of the cousin marriage will have the trait.



77. Mode of inheritance _____

Ans: Recessive

Difficulty: 1

78. Prob. 1×6 _____

Ans: $1/3$

Difficulty: 4

79. Prob. 3×7 _____

Ans: $1/6$

Difficulty: 4

80. In some plants, a purple pigment is synthesized from a colorless precursor. In a cross between two plants, one purple and the other colorless produced an F_1 that was purple. The F_2 produced from the F_1 had 775 purple, 200 Red and 65 colorless. What is the genotype of the parents?

Purple parent	Colorless parent

Ans: The ratio is a 12:3:1; Parents: $AABB \times aabb$

Difficulty: 3

81. Short hair in rabbits is governed by a dominant gene (l^+) and long hair by its recessive allele (l). Black hair results from the action of the dominant gene (b^+) and brown hair is from its allele (b). Give the genotypic and the corresponding phenotypic ratios of the F_1 from a cross of a female rabbit with brown hair and male rabbit with long hair. This assumes female is homozygous for short hair and male is homozygous for black hair.

Ans: Genotype Phenotype
 l^+l b^+b

Difficulty: 1

82. Short hair in rabbits is governed by a dominant gene (l^+) and long hair by its recessive allele (l). Black hair results from the action of the dominant gene (b^+) and brown hair is from its allele (b). Give the genotypic and the corresponding phenotypic ratios of the F_2 from a cross of a female rabbit with brown hair and male rabbit with long hair. This assumes female is homozygous for short hair and male is homozygous for black hair.

Ans: #	<u>Genotype</u>	<u>Phenotype</u>
1	$l^+l^+ b^+b^+$	Short Black
2	$l^+l b^+b^+$	Short Black
2	$l^+l^+ b^+b$	Short Black
4	$l^+l b^+b$	Short Black
1	$l^+l^+ bb$	Short Brown
2	$l^+l bb$	Short Brown
1	$ll b^+b^+$	Long Black
2	$ll b^+b$	Long Black
1	$llbb$	Long Brown

Difficulty: 3

83. Stem color of tomato plants is known to be under the genetic control of at least one pair of alleles such that A^- results in the production of anthocyanin pigment (purple stem). The recessive genotype aa lacks this pigment and hence is green. The production of two locules in the tomato fruit is controlled by the dominant allele M and multiple locules is determined by mm . Give the genotypic and phenotypic ratios of the F_1 a cross between an inbred tomato plant that has a purple stem and fruit with two locules crossed to a tomato plant that have fruit with multiple locules and a green stem.

Ans: <u>Genotype</u>	<u>Phenotype</u>
$AaMm$	Purple & 2 locules

Difficulty: 2

84. Stem color of tomato plants is known to be under the genetic control of at least one pair of alleles such that A- results in the production of anthocyanin pigment (purple stem). The recessive genotype aa lacks this pigment and hence is green. The production of two locules in the tomato fruit is controlled by the dominant allele M and multiple locules is determined by mm. Give the genotypic and phenotypic ratios of the F₂ from a cross between an inbred tomato plant that has a purple stem and fruit with two locules crossed to a tomato plant that have fruit with multiple locules and a green stem.

Ans: #	Genotype	Phenotype
1	AAMM	Purple, 2 locules
2	AaMM	Purple, 2 locules
2	AAMm	Purple, 2 locules
4	AaMm	Purple, 2 locules
1	aaMM	Green, 2 locules
2	aaMm	Green, 2 locules
1	Aamm	Purple, Multi loc
2	AAMm	Purple, Multi loc
1	aamm	Green, Multi loc

Difficulty: 3

85. What does a diamond symbol Δ in a pedigree indicate?

Ans: sex unspecified

Difficulty: 2

86. What is does a vertical pattern of inheritance in a pedigree likely indicate?

Ans: rare dominant trait

Difficulty: 2

87. Calculate the probability of the production of a homozygous recessive genotype for the following cross: AaBbccddEeFf $\times$ AaBbCcddEeFf

Ans: $\frac{1}{4} \times \frac{1}{4} \times \frac{1}{2} \times 1 \times \frac{1}{4} \times \frac{1}{4}$

Difficulty: 3

88. Calculate the probability of either all dominant or all recessive genotypes for the alleles A, B, E, and F in the following cross: AaBbccddEeFf $\times$ AaBbCcddEeFf

Ans: $(\frac{3}{4} \times \frac{3}{4} \times \frac{3}{4} \times \frac{3}{4}) + (\frac{1}{4} \times \frac{1}{4} \times \frac{1}{4} \times \frac{1}{4})$

Difficulty: 4

89. What are the four general themes that have arisen from Mendel's work?

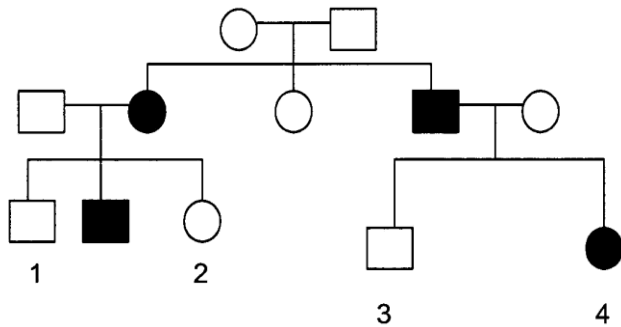
Ans: Variation, as expressed in alternative forms of a trait, is widespread in nature.

Observable variation is essential for following inheritance of traits. Variation is not distributed by chance alone but is inherited according to the genetic tenet that "like begets like." Mendel's laws apply to all sexually reproducing organisms.

Difficulty: 4

Diagram Problems

What are the possible genotypes of persons 1, 2, 3 and 4?



90. Person 1

Ans: Aa

Difficulty: 2

91. Person 2

Ans: Aa

Difficulty: 2

92. Person 3

Ans: Aa

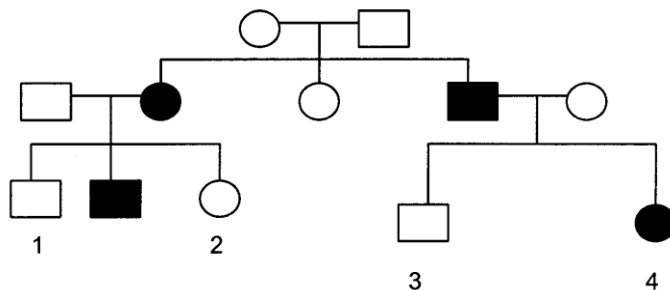
Difficulty: 2

93. Person 4

Ans: aa

Difficulty: 2

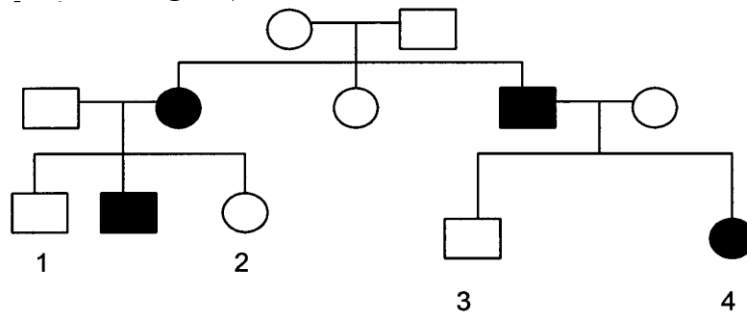
94. Below is a pedigree of a human genetic disease in which stricken individuals are solid-colored. Apply the laws of probability and calculate the probability the offspring of the cousin marriages a) 2 × 3 will have the disease.



Ans: The trait is a recessive trait: $\frac{1}{4}$

Difficulty: 2

95. Below is a pedigree of a human genetic disease in which stricken individuals are solid colored. Apply the laws of probability and calculate the probability the offspring of the cousin marriages a) 1×4



Ans: The trait is a recessive trait; $\frac{1}{2}$

Difficulty: 1

Experimental Design and Interpretation of Data

96. A youngster has dozens of pet mice and asks you why their coat colors are so different. He explains that his favorite color is black with white patches and wonders how he can get more of them, yet his favorite mice are actually the “nice” white ones who are gentler than the active nippy black ones. You decide to give a simple genetics lesson. Help the youngster set up an artificial selection for “nice,” black mice with white spots, including an indication of expected results and an interpretation of the data.

Ans: Breed male black mice with female white mice and male white mice with female black mice. Interbreed the offspring. Test all black mice with white spots for gentle behavior and breed the gentlest males to the gentlest females. If the genes for coat color and gentle behavior are unlinked, not closely linked, or not the same gene, this experimental design will result in the production of gentle mice that are black with white spots. If selection pressure is continued for numerous generations, the genes will become fixed in the population and gentle black mice with white spots will always result.

Difficulty: 4

97. As an owner of an orchard you realize that the selective breeding of apple trees to produce the most beautiful red apples have left customers with a displeasure toward the now bland tasting beautiful apples. The results indicate what about the two traits? How would you as an orchard owner fix the problem for the long-term?

Ans: The genes for taste and red color are not linked. To fix the beautiful, tasteless, apple problem, a rederivation of the apples is necessary. It will be necessary to cross the beautiful red apple tree with apples with outstanding taste. Each generation should be observed for taste and color with the most tasty and colorful being crossed to each other. As you might imagine, fixing the problem in your orchard will take a very long time.

Difficulty: 4

98. You are out on a nature walk up in the mountains and you find a pretty wild flower in the lower altitude that is short and bushy with small fragrant, bright purple flowers and in the higher altitude you find what seems to be the same plant yet it is tall and sparse with larger flowers of the same color and fragrance. a) Set up an experiment to test the hypothesis that the plants are different due to genetic but not environmental influences. b) Is it possible to tell if both genetic and environmental effects occur?

Ans: a) Obtain several specimens from each location. Plant seeds of both types of plants in both low and high altitude locations. Observe the offspring. If the offspring look the same as their parental stock then the differences are simply genetic in nature. If the offspring look short and bushy with small fragrant, bright purple flowers in the lower altitude and tall and sparse with larger flowers of the same color and fragrance in the higher altitude then the differences are due to simple environmental influences. b) Yes, a combination of the traits would indicate that both environmental and genetic influences play a role in the differences you have identified.

Difficulty: 3

99. You wish to know the genotype of some carrot plants that you have grown in your garden so that you might grow more of them. They have redish-orange flesh, are sweet in taste, long in root and short in leaf. Using classical genetic techniques how would you determine the genotype?

Ans: You need to determine the recessive nature of each trait. Set up crosses between redish-orange, sweet tasting flesh, long in root and short in leaf and true orange, plain tasting, short in root and long in leaf to determine each dominant trait. Then create a tester plant that is recessive for all four traits. Cross your favorite carrot plants with the tester and observe the offspring. The traits shown in the offspring are indicative of the genotype of your original carrot plant.

Difficulty: 4

100. You are talking to your father about your relatives and he shares with you that there is a late-onset disease that seems to run in his family. What could you do to determine your probability of having this late-onset disease?

Ans: Create a pedigree of your family tree for the late-onset disease going back at least three but as many generations as possible. Based on your pedigree, you will need to determine whether the trait is recessive or dominant, autosomal or X-linked. Use the product rule to determine the probability of your having inherited the trait. Keep in mind that individuals not old enough to show the trait should be diagramed as unknowns on your pedigree and your probability of inheriting the disease may depend on whether that given individual has the trait.

Difficulty: 2